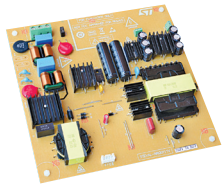


gy-efficiency and standby regulations. The board provides a 12V/4A regulated output to power the television's controller and audio subsystem and a 65V/2.5A output for LED backlighting. No coding is needed as the digital control



algorithms are permanently programmed in the controller's ROM and the board comes with a complete set of sample parameters stored in programmable non-volatile memory.

STMicroelectronics
www.st.com

High-current DC-DC converter

Advanced Energy has introduced the Artesyn LGA110D, a 110-ampere, non-isolated digital point-of-load DC-DC converter. It delivers a higher current rating at 30% higher power conversion density than any DC-DC converter in its



class, making it a superior solution for a broad range of applications in the industrial, medical, telecom, and IT infrastructure markets. In addition to its superior current density (220 amps per square inch), the LGA110D features two independently controlled 55-amp/175-watt outputs, which means fewer power converters are needed in multiple channel or high current applications.

Advanced Energy
www.advancedenergy.com

Compact power module

The BMR310 from Flex Power Modules is built on the proprietary zero voltage switching switched capacitor converter technology of Infineon Technologies AG and achieves efficiency of over 98% at



half load. It can deliver power up to 860W continuous in a compact package. The module operates over an input voltage range of 40V to 60V and delivers an unregulated output voltage of 10V to 15V. Data centres and high-performance computing applications need 48V power

distribution to reduce distribution currents and power losses. This is achieved through capacitive energy transfer with soft switching of the power MOSFETs.

Flex Power Modules
www.flexpowermodules.com

Small sensor board

Arduino Pro and Bosch Sensortec have partnered to set a new standard for smart sensing solutions. The Nicla Sense ME features the latest generation Bosch sensors technology on the smallest form factor yet, to bring sensing and intelligence at the edge. Featuring a 9DoF smart motion sensor and a 4DoF environ-



mental sensor with AI capabilities, the co-branded board allows for a wide range of applications to address the different segments in the IoT market, while maintaining simplicity of integration and scalability. The board comes with Bosch Sensortec's BHI260AP AI sensor system with integrated motion sensor, BMM150 magnetometer, BMP390 pressure sensor, and the BME688 4-in-1 gas sensor with AI and integrated high-linearity and high-accuracy pressure, humidity, and temperature sensors.

Arduino
www.arduino.cc

AI thermal sensing solution

Arrow Electronics has launched a thermal sensing solution that is powered by STMicroelectronics' STM32CUBE AI. The integrated solution is designed to help engineers and product designers rapidly combine AI and thermal sensing technologies for



developing smart, reliable, and affordable health-monitoring devices. This integrated solution optimises bill-of-materials and simplifies hardware and software integration, making it a good choice to deploy for temperature-screening devices and a wide variety of other consumer-grade and healthcare applications. The AI-powered thermal sensing solution can achieve a quick and accurate temperature screening with multiple

individuals screened simultaneously.

Arrow Electronics
www.arrow.com

100W DC-DC converter

The TEP 100UIR from Traco Power is a series of high-performance 100W DC-DC converters with an ultra-wide 12:1 input voltage range. These come in a compact, fully encapsulated quarter-brick pack-



age for highest reliability. The ultra-wide input voltage range lets system builders reduce the number of design channels where otherwise multiple input voltage configurations would be required. Thanks to its dedicated holdup circuit, the TEP 100UIR complies with extended holdup-time requirements without the need for bulky capacitors. It meets certifications EN 50155 for rolling stock applications, EN 61373 for mechanical shock and vibration, and EN 45545-2 for fire behaviour.

Traco Power
www.tracopower.com

V2X hardware security module

Advancements in the number of communication interfaces in vehicles is growing. However, system vulnerability due to rising cyberattacks poses a challenge



to the safety of passengers and the protection of their data. To provide a high level of protection against such threats, Infineon Technologies is launching the SLS37 V2X hardware security module, a plug-and-play security solution for vehicle to everything communication (V2X). It is based on a highly secured, tamper resistant microcontroller tailored to the security needs in V2X applications within telematics control units. It protects the integrity and authenticity of messages, as well as the privacy of the sender.

Infineon Technologies AG
www.infineon.com

Next-gen processor platform

congatec has introduced the conga-QMX8-Plus, a Qseven module based